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Fracture Height and Barrier Identification in a Conventional Formation: Introducing a Fracture Barrier Index and its Implications to Fracture Simulation and Well Productivity

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Abstract

The AMN Formation is a sandstone reservoir and the primary target in the FKR field development in Sultanate of Oman. Over 12 wells have been drilled in recent years, accompanied by numerous studies aimed at understanding its production potential and fracability. The processes governing hydraulic fracture geometries in the AMN Formation are continuously being assessed, particularly due to the complexity introduced by fine-scale dispersion of formation properties and their vertical variability.

Fracture height is a critical parameter for several decisions, including the depth of penetration into the formation, the number of stages (which serve as inputs to production models), and calibration of fracture simulations. Its significance is further emphasized in reservoirs characterized by multiple fluid phases: free gas, trapped gas, and water zones.

In this paper, we examine the intrinsic fine-scale features from open-hole data that influence fracture height growth, with a particular emphasis on moving beyond the common assumption of a homogeneous reservoir. Additionally, we explore the broader implications of transitioning from a homogeneous to an anisotropic reservoir model for fracture simulation and, ultimately, well productivity. We focus on the following factors: sandstone density, principal stress, geomechanical properties, and laminations with lithological contrasts.

The AMN Formation is heterogeneous and anisotropic under a normal stress regime. This anisotropy was derived from correlating open-hole features with actual fracture data from previous wells. The resulting 1D anisotropic geomechanical model significantly impacts fracture height and, consequently, well production performance. This conclusion is supported by actual fracture job data.

As observed, the computation of 1D geomechanical models using both low- and high-resolution inputs affects fracture height simulation. In this paper, we present an example to highlight these differences and the consequences of the averaging process used during stage selection in fracture simulation, which in turn influences the resulting fracture height growth.

While the presence of fine-scale reservoir features in the AMN Formation has been previously documented, their impact on fracture height has not been thoroughly examined. This study extends the investigation across the field to enhance well productivity by effectively managing fracture height to cover the entire pay zone.

The inherent challenges in detecting, characterizing, and accurately modeling these fine-scale features—and simulating their interaction with hydraulic fractures—are addressed through a developed workflow. This workflow is designed to characterize these features and incorporate them as inputs for fracture simulation. Additionally, we discuss the field data that contribute to understanding and calibrating fracture height and production allocation. The paper concludes with a summary of the effects, their influence on fracture height, and their implications for fracture simulation and well productivity.

Introduction

From late 2000 to early 2025, more than 300 hydraulic propped fracturing stages were performed by Petroleum Development Oman (PDO) within the Gas Assets and Exploration department across various fields targeting the AMN Formation (Figure 1). These concession fields are located within the Fahud and Ghaba Salt Basins, with bottom-hole temperatures (BHT) ranging from 156°C to 173°C and are characterized by tight sandstone reservoirs. The typical well completion is a perforated vertical cased hole, although some fields feature diverse completions, including cased-hole horizontal wells and cased-hole multi-cluster configurations.

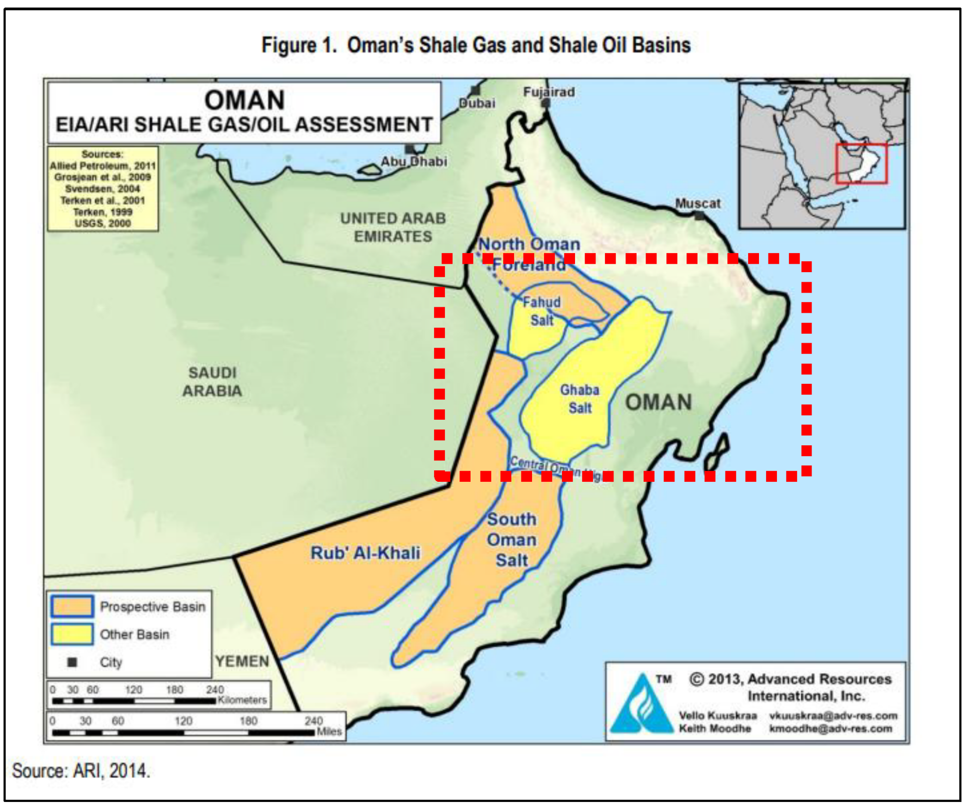

Figure 1—North (red dash) fracturing operation area within Sultanate of Oman petroleum basin as ARI basin map

The AMN Formation, Cambrian in age, is part of the Haima Group. Current interpretations suggest that it comprises a mixture of fluvial, sabkha, and aeolian sediments deposited within a low-relief extradunal sabkha sandsheet to margin setting. Despite the common misconception that the reservoir consists of a thick, homogeneous sand body with a single hydraulic unit, it is stratigraphically divided into distinct geological sandstone sub-units—S-30, S-20, and S-10. These sub-units exhibit inherited geological heterogeneity and mechanical variation, which has been evidenced by the difficulty in achieving uniform hydraulic propped fracturing. This challenge has impacted success rates in terms of fracture placement, production, and fracture height, even among wells with similar Petrophysical properties and lithology.

The variability of the AMN Formation across different basins has introduced multiple challenges for hydraulic fracturing treatments, necessitating fit-for-purpose strategies and technical solutions tailored to each well's characteristics and issues. Fracture height growth is a critical factor in the design and effectiveness of hydraulic stimulation. Although open-hole logs show no clear stress barriers, results from all jobs traced with RA tracers indicate very confined treatments, with no upward or downward fracture propagation. This has led to inefficient proppant placement and suboptimal production.

Various fracture simulation modeling software have been used, all of which assume uniform geomechanical behavior within and across the pay zone layers. However, field experience increasingly suggests that local heterogeneity plays a dominant role in fracture propagation.

To improve production, hydrocarbon recovery, and hydraulic fracturing design, the Fracture Barrier Index (FBI) has evolved into a multidisciplinary predictive tool that integrates geological, geomechanical, and regional insights. The core components of the enhanced FBI include fine-scale geology, regional tectonics, in-situ stress modeling, and analogous formations. This enhanced approach transforms AMN's uncorrelatable heterogeneity into a quantifiable containment predictor.

Moreover, the focus of these studies originated from independent teams and was later consolidated through collaboration between fracturing subject matter experts (SMEs) and the rock mechanics team. The next section will discuss the challenges encountered and the studies conducted to develop solutions, based on experience from the broader fracturing campaign in the AMN Formation and specifically within the FKR field.

Problems Identification and Studies Area

From a geological perspective, the AMN reservoir is located in the southern part of the internal region of the Sultanate's ophiolite belt (Figure 2), characterized predominantly by tight rock formations. Due to its proximity to the subduction zone between the Arabian and Eurasian plates, tectonic activity has significantly influenced the lithostatic stress regime in the area. As a result, the conventional orientation of the maximum horizontal stress is typically within the 40–60° NESW direction, unless modified by localized tectonic elements.

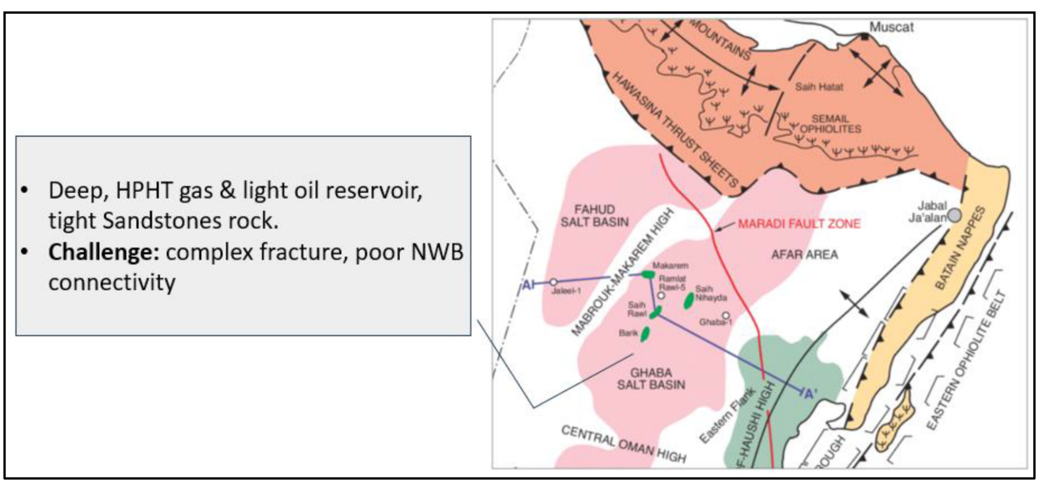

Figure 2—Oil and Gas North within Ghaba Salt basin; reservoir characteristic and common challenges

In the northern region, most hydraulic fracturing activity occurs within the high-pressure, high-temperature (HPHT) tight rock fields of the Ghaba Salt Basin, where stress regimes and fracture behavior are predominantly controlled by regional tectonics and rock fabric.

In the deep HPHT reservoirs of the Ghaba Salt Basin, hydraulic fracturing designs typically incorporate higher proppant concentrations to compensate for conductivity loss caused by high closure stress post-fracturing, as well as damage from high-temperature (HT) fracturing fluids with elevated polymer and chemical loadings. However, placing larger proppant sizes at higher concentrations in such high-stress formations requires excellent near-wellbore (NWB) connectivity. This necessitates larger perforation entrance diameters and higher shot densities to ensure effective proppant transport and placement.

Higher closure stress also reduces stress anisotropy between the minimum and maximum horizontal stresses, which can lead to the opening of fissures or the development of T-shaped fractures. Both scenarios introduce secondary leak-off behavior, posing a risk to proppant placement once these features are activated.

Historically, hydraulic fracture modeling relied on simplified assumptions of rock brittleness, typically calculated using only Poisson's ratio and Young's modulus. However, field experience—particularly from frac jobs in the AMN Formation across various gas fields, including the FKR field—has shown that these parameters alone are insufficient to accurately predict fracture-ability. In several AMN wells, fracturing attempts failed despite favorable brittleness indices, while others succeeded under less promising conditions.

The understanding of frac propagation has been reshaped thorough investigation done for the fracture height and the barrier identification in the conventional gas formation including natural fractures, stress orientation and local variations in ductility. The concept of a hydraulic barrier has advanced beyond the mechanical contrast and included Multi-Variable Barrier Model that was able to explain the variation within "geologically similar" but with different HF metrics.

The Multi-Variable Barrier Model accounts for several fine-scale geological and geomechanical features that influence fracture propagation and containment in the AMN Formation:

- Differences in Horizontal Stress Magnitude (ΔShmin): Variations in the minimum horizontal stress across layers can create stress contrasts that either promote or restrict fracture height growth.
- Rock Fabric Orientation and Texture: Cross-bedded intervals with dip angles between 15° and 40° tend to deflect fractures laterally, altering their height propagation path.
- Laminated intervals, on the other hand, tend to force planar fracture growth but restrict vertical propagation, acting as natural barriers.
- Compositional Laminations (e.g., High TOC or Carbonate Streaks): Thin carbonate-cemented layers (typically <30 cm thick) can act as localized barriers. These are often identified by subtle increases in density and slight reductions in porosity on well logs.
- Pre-Existing Fracture Networks: Natural fractures can either enhance or hinder hydraulic fracture propagation depending on their orientation, connectivity, and interaction with the induced fracture.
- Fine-Scale Permeability Streaks: Alternating high- and low-permeability streaks within the reservoir can influence fluid leak-off behavior and fracture containment, especially in laminated or heterogeneous zones.

Since this approach demonstrated a pragmatic solution to AMN heterogeneity, it was applied to the FKR Field. The subject well exhibited a porosity range of 3–16%, permeability between 0.001–1 mD, and unconfined compressive strength (UCS) values ranging from 100–250 MPa. Previous frac jobs revealed that reservoir strength heterogeneity had a greater influence on fracture behavior than stress contrast. One of the key outcomes of the Multi-Variable Barrier Model was the recognition that fine-scale features predominantly control fracture behavior—particularly fracture height growth.

A detailed analysis of existing wells, using open-hole data integrated with core descriptions, showed that these fine-scale features are laterally correlated across the field, although their vertical presence varies from well to well.

Additionally, a previous study on the same formation emphasized that multi-rate temperature and acoustic logs significantly enhance the understanding of vertical fluid migration and help confirm fracture containment across expected barriers—aligning with the current understanding of the formation.

A joint collaboration between PDO's Petrophysical Gas Asset team, hydraulic fracturing SMEs, and service company experts further categorized the challenges of fracture confinement and barrier identification into two key study phases:

1. **Pre-Fracturing Phase** – Focused on fracture design, stage and perforation selection, and placement strategy.
2. **Post-Fracturing Phase** – Focused on flowback analysis and RA tracer interpretation.

To optimize team resources and study focus, the investigation was narrowed to the **AMN S-30 sub-unit**, leveraging analog field data and concentrating on 11 hydraulically fractured wells within the FKR field.

Furthermore, the team incorporated insights from an alternative geomechanical study conducted on an offset well in the same field. The barrier strategy for that well was revised post-fracturing, as the actual barrier strength was found to be greater than initially anticipated (Figure 3).

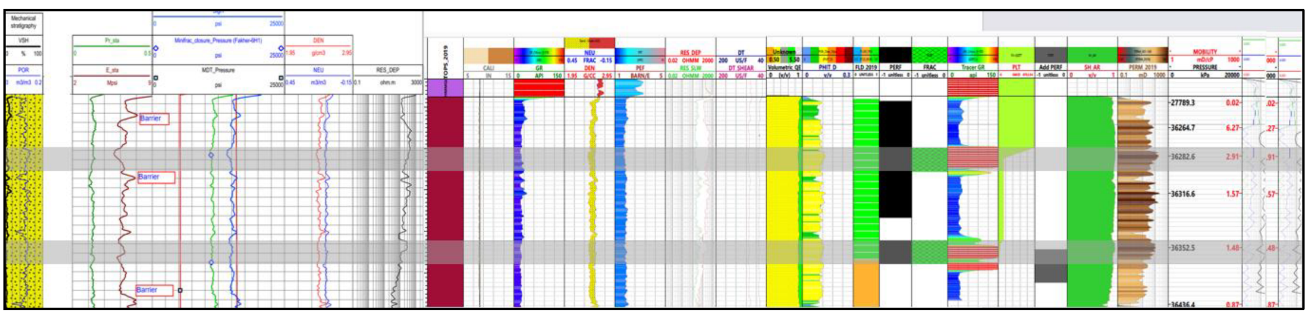

Figure 3—Analog well AMN S-30; pre-frac geo-mechanical study and post frac frac propagation

Therefore, the offset well was pressure-matched after the fracturing operation to align the modeled fracture height and leak-off behavior, enabling a more accurate stress analysis for the current well. A strong correlation was observed between the calculated Fracture Barrier Index (FBI) and the pressure-matching model results (Figure 4 and Figure 5).

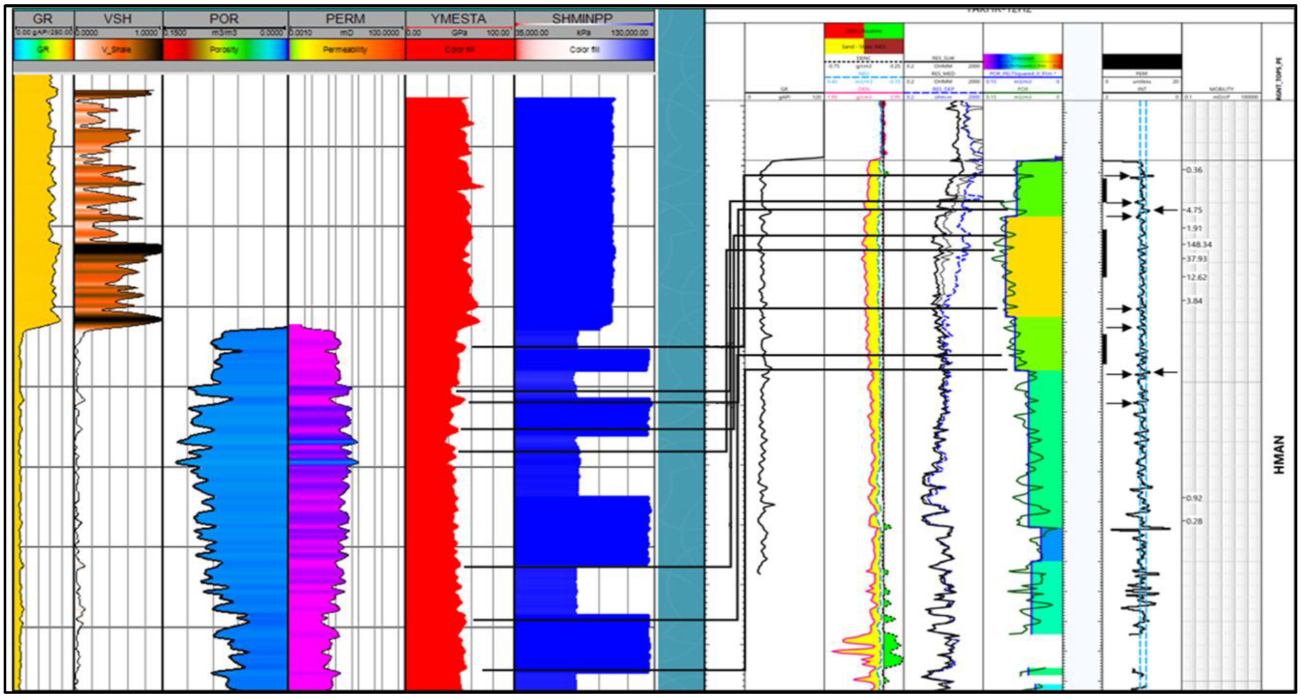

Figure 4—Offset well AMN S-30 barriers are matched with calculated FBI for post frac pressure matching.

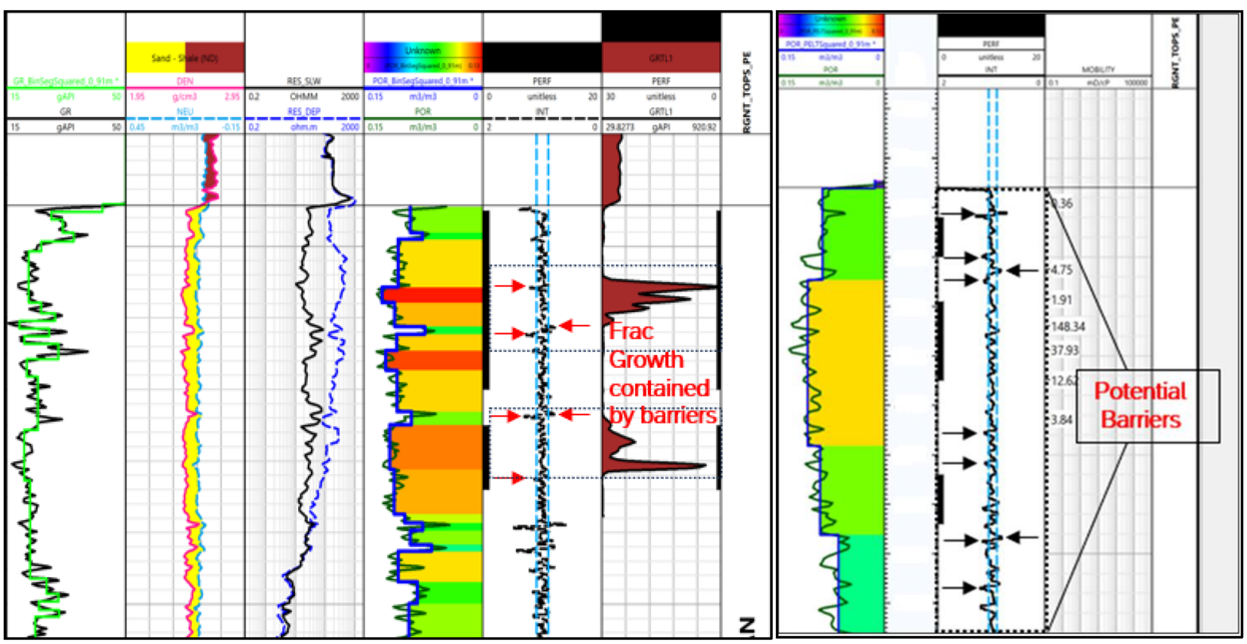

Figure 5—Good matching of the potential barriers from calculated FBI with actual tracer results and post frac pressure matching.

Methodology

FBI (Fracture Barrier Index) is a method used in hydraulic fracturing planning to identify and predict geological barriers that can constrain fracture growth. It is basically integrating multiple geological, petrophysical and geomechanical properties to guide the design of hydraulic fracturing, especially where fracture growth is critical parameters like in the case of multiple fluid fill columns.

Below is the technical breakdown that summarizes the FBI approach and its rationality.

1. Data Integration & Preparation:
   a. Data Collection:
      i. Open Hole Logs: Density, Sonic and Resistivity Log data for all candidates' wells.
      ii. Core Description: Detailed core description including lithology and texture analysis from core wells.
      iii. Tracer Logs: Collecting all tracer logs post frac for all frac wells in FKR Field to ass the fracture height growth.
      iv. Memory Production Logging Tool (MPLT): To validate the flow contribution and to confirm the barrier effectiveness.
      v. Geomechanics Data: Collecting in-situ stress profile, UCS from sonic log and 1D MEM.
   b. Data Quality Control: The objective from this is to check log consistency, depth shift and tools issues. Also, normalization was considered as pre-request since the model will be applied for multiple wells.
2. Feature Extraction & Smoothing:
   a. Smoothing Geological Noise:
      i. Applying Gaussian filters to density logs to remove any noise and focus on the major geological transitions.
   b. Detecting Abrupt Changes (Change Detection):
      i. Calculate the first derivative of porosity with respect to depth
      ii. Flag intervals where this change exceeds a set of thresholds indicating potential barriers
   c. Cross-Referencing with Core Data:
      i. Validate log derived features against core deception to confirm geological origin.
3. Parameters Calculation & Threshold Calibration:
   a. Key Parameters:
      i. Porosity Gradient: Indicate abrupt lithological changes
      ii. UCS: Derivate from Sonic logs where high UCS (>160 MPa) signifies strong fracture resistance rocks.
      iii. Permeability: Based on core transform where low perm (<0.1mD) typically marks barriers
   b. Tracer Log Calibration:
      i. Compare flagged barriers from logs with actual fracture height from tracer logs
      ii. Iteratively adjust gradient and other parameters threshold to achieve 90% agreement between predicted and observed barriers.
4. Fracture Barriers Index (FBI) Scoring Logic:
   a. Assigning FBI score under each depth using the following logic:
      i. FBI= 1 (Barrier): $\Delta\phi/\Delta z \geq 4\%/\text{ft}$, UCS > 160 MPa, k < 0.1 mD
      ii. FBI=0 (Target): $\Delta\phi/\Delta z < 2\%/\text{ft}$, UCS < 140 MPa, k > 0.1 mD
      iii. 0 < FBI < 1 (Transition): It will be a linear interpolation for intermediate values
   b. Generating FBI Curve: Plotting the FBI score as continues curve, highlighting the high -probability barriers and target intervals.
5. Application in Fracture Design & Execution:

   a. Perforation & stage selection
      i. Target perforation Intervals: FBI < 0.3, this is indicating low risk of barriers interference.
      ii. Avoid: where FBI > 0.7, as normally these are strong low perm barriers
      iii. Transition Zones: consider using specialized type of fluids if perforation is required
   b. Facture treatment optimization:
      i. Adjust fluid parameters based on FBI profile.
      ii. For zones where they are near barriers, use different type of fluids to minimize the screen out.
6. Post-Frac Validation & Continuous Improvement:
   a. Run tracer logs/MPLT to measure actual fracture height and their contribution
   b. Use this data to recalibrate thresholds and improve FBI model for future wells

depth specific barrier plot example is shown below (Figure 6) with the FBI matrix and the detailed FBI flowchart as below in Figure 7.

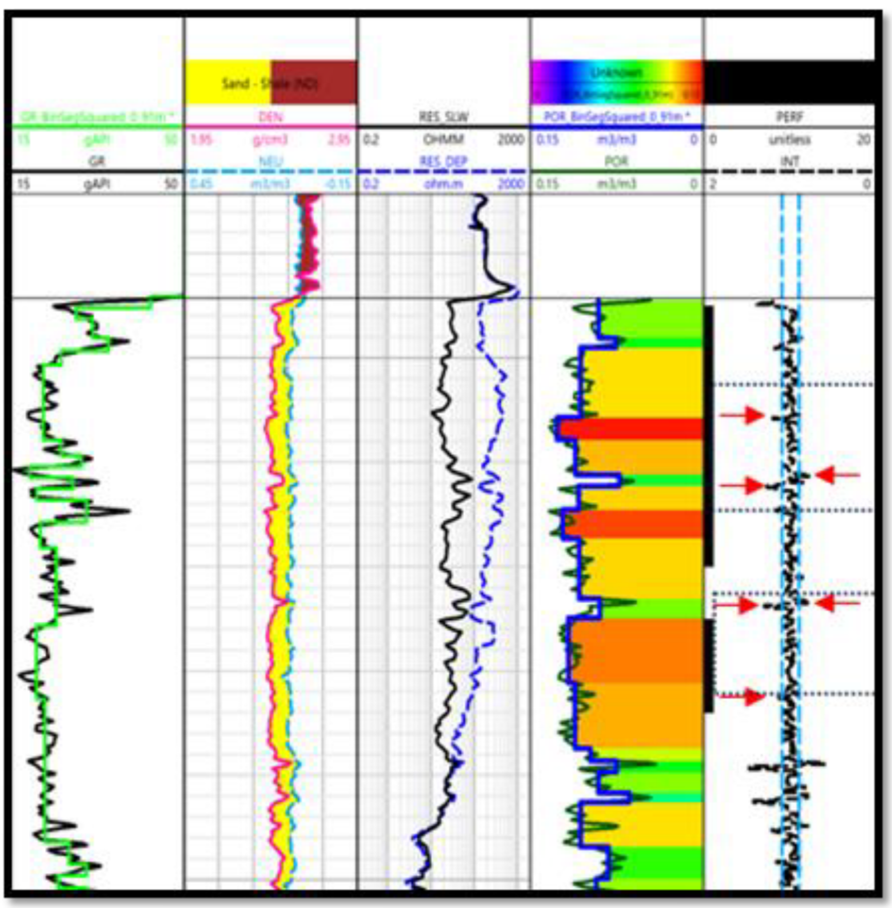

Figure 6—AMN formation frac propagation; RA tracer proven the fracture contained

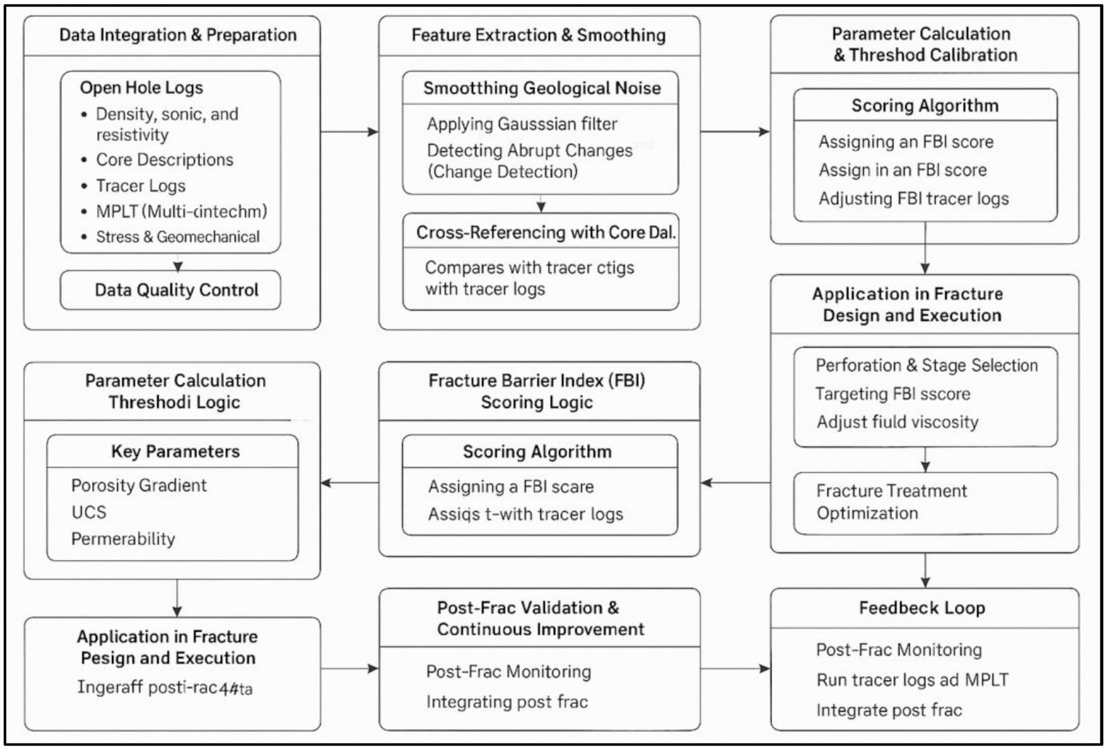

Figure 7—FBI flowchart summarizing the steps used for FBI that we have implemented

Case Study: AMN S-30 Formation at FKR Field

Over 30 hydraulic fracturing attempts have been analyzed across the AMN Formation in the FKR Field, specifically targeting the S-30 sub-unit. Most of these treatments exhibited confined fracture propagation across the layers, regardless of minor variations in horizontal stress magnitude. This observation reinforces the understanding that even small differences in horizontal stress can act as effective barriers during fracturing operations—the greater the horizontal stress contrast, the more likely it is to restrict fracture growth.

A case study from one of the FKR wells demonstrates the practical application of the Fracture Barrier Index (FBI) in the AMN Formation. Two stages were planned and executed in this well using a crosslinked fluid system combined with a degradable fluid loss additive during the pad stage. This careful fluid selection helped control fracture dimensions, maintain the desired geometry, and mitigate fluid loss.

Two types of proppants were used in this well: 30/50 High Strength Proppant (HSP) and a Relative Permeability Modifier (RPM) proppant. The RPM proppant was specifically chosen to address the known condensate banking issue in the field, enhancing hydrocarbon flow. These proppants were selected based on the field's unique geological and petrophysical characteristics, ensuring optimal performance and effectiveness of the fracturing treatment.

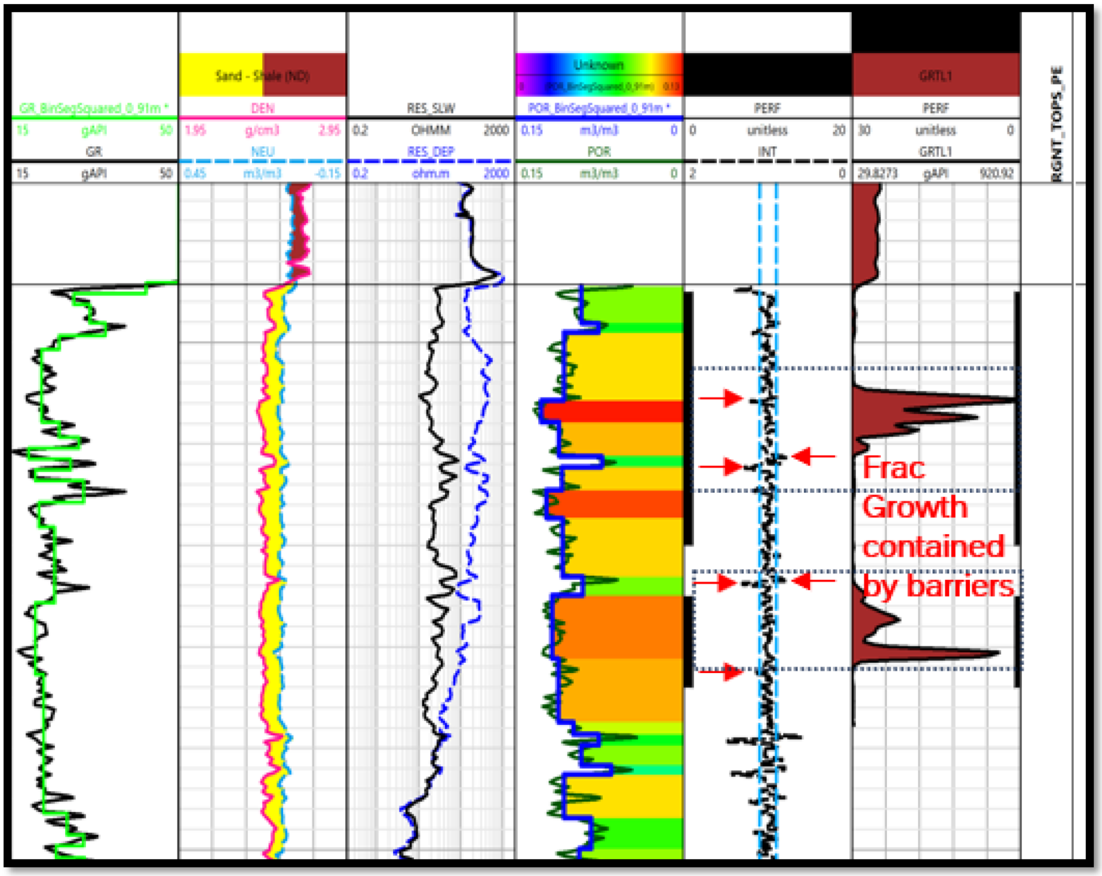

Figure 8—FBI log of FKR-12 along with the barriers model and tracer log post frac

Pressure matching was conducted along with tracer logging on this well as per below:

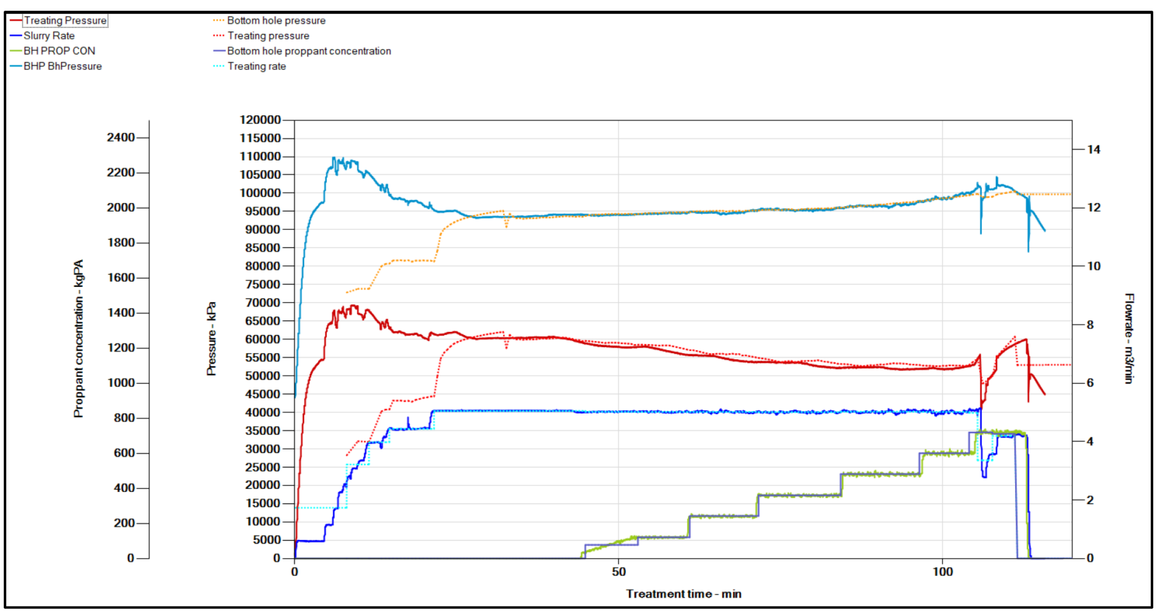

Figure 9—Stage-1 AMN S-30; Post frac pressure matching showing a good matching of the treatment parameters.

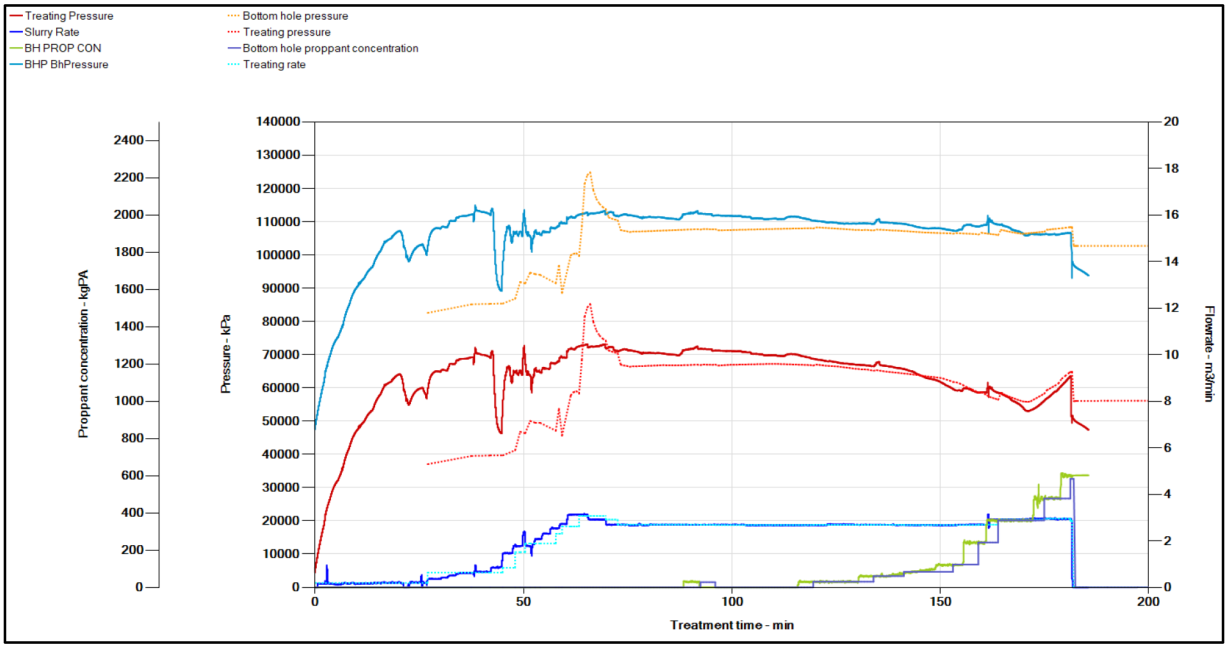

Figure 10—Stage-2 AMN S-30; Post frac pressure matching showing a reasonable matching of the treatment parameters. A clear indication of a broken barrier (big reduction in treating pressure) right at the beginning of the treatment.

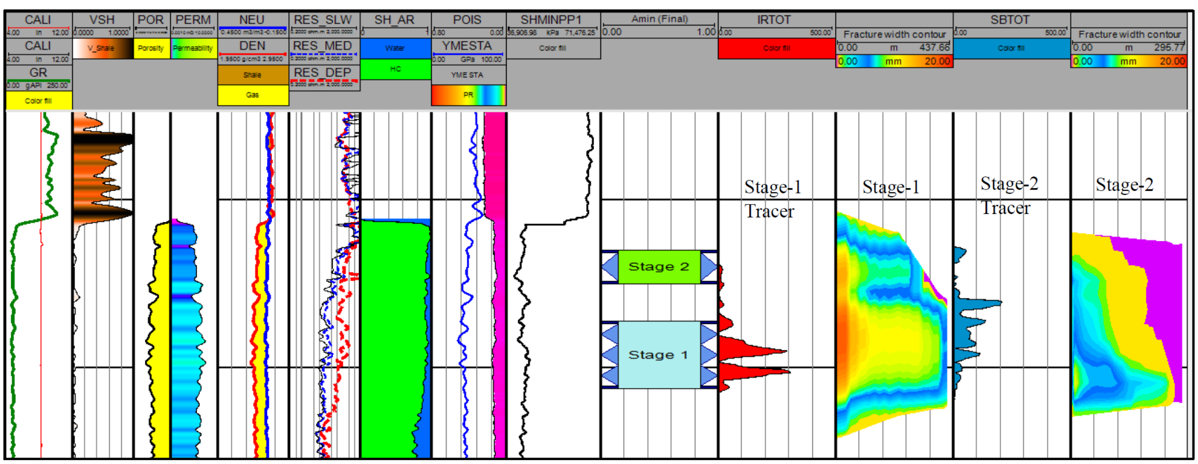

Figure 11—AMN S-30; Post frac pressure matching of both stages compared with tracer logs. The analysis showed the fracture of stage-1 was mostly confined to the interval, covering the entire perforation interval while stage-2 shows poor confined fracture within the interval with majority of the proppant placement is occurring between the 2 stages below stage-2 interval. It also shows good correlation with the analysis of the tracers for both stages.

During both stages of fracturing, the fracture height propagated sufficiently to connect both fractures together, enhancing the overall connectivity. To manage and optimize the fracture geometry.

Stage	Perforation length (meters)	Fracture Height (meters)
Stage-1	8	27
Stage-2	4	22

Observed Barriers

Gas saturation log saturation calculation from the open hole log from well to another confirmed the containment of fractures by observing desaturation anomalies across the suspected barriers.

Fracture Simulation Alignment

FBI values were validated against minifrac and tracer from main frac results, showing strong correlation between high FBI zones and elevated breakdown pressures. Application of the FBI allowed for more realistic fracture modeling, improving the accuracy of predicted fracture height and volume.

Implications for Stimulation Design and Well Productivity

Hydraulic fracturing has proven to be an effective method for enhancing production from the AMN Formation. As such, zonal coverage across the pay zone plays a critical role in determining both the initial production rate and overall recovery. In this context, the Fracture Barrier Index (FBI) can serve multiple roles:

- Frac Stage Optimization: Prioritizing intervals with low FBI scores to maximize proppant efficiency and ensure effective fracture propagation.
- Barrier Mapping: Identifying high-FBI layers to avoid or redesign treatments, reducing the risk of fracture containment and screen-outs.

By integrating FBI with seismic-derived stress models and quantitative interpretation (QI)-based ductile markers, operators can more confidently forecast fracture growth and placement. This integrated approach directly enhances well performance and improves economic outcomes.

Conclusion and Recommendation

Post-hydraulic propped fracturing challenges-particularly confined fracture growth and limited zonal coverage-have been thoroughly studied. In this context, the use of radioactive tracers proved successful in validating the Fracture Barrier Index (FBI) methodology. The results confirm that FBI is a practical and applicable approach for formations like AMN. The key findings include:

- Fracture height and containment in conventional formations are governed by a complex interplay of rock properties, stress field variations, and structural heterogeneities.
- The Fracture Barrier Index (FBI) consolidates key geological and geomechanical variables into a predictive framework for identifying intervals resistant to fracture propagation.
- Field data from the AMN Formation validate the FBI concept, demonstrating its potential to improve fracture simulation accuracy and stage design.
- Cross-comparison with other fields and published studies highlights the broader applicability of incorporating tectonic, mechanical, and microstructural features into fracture modeling workflows.

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Glossary

FBI	Fracture Barrier Index
HPHT	High Pressure High Temperature

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